READING 2.3 · SHIP AND OCEAN SYSTEMS DESIGN

The Admiralty Law

Power, speed and displacement for one hull form: the physics, the deduction, its limits, parent ships and a worked example

Papanikolaou (2026) §1.3, §2.3, §2.14 · Session W2.3 · extended reading after Lecture 2.3a, not for evaluation

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Where the Admiralty Law Comes From

1

$$R_T = \tfrac{1}{2}\,\rho\,S\,V^2\,C_T \qquad P_E = R_T\,V, \qquad P_B = \frac{P_E}{\eta_D\,\eta_S}$$

Resistance; then effective and brake power through the propulsive and shaft efficiencies

2

$$S = C_S\sqrt{\nabla L} = C_S\sqrt{L/\nabla^{1/3}}\;\nabla^{2/3} = k_S\,\nabla^{2/3}, \qquad \Delta = \rho\,\nabla$$

Taylor (Eq. 2.8)[1]: one hull shape fixes CS and L/∇ to the 1/3, so kS is constant

3

$$P_B = \frac{R_T\,V}{\eta_D\,\eta_S} = \frac{\tfrac{1}{2}\,\rho\,k_S\,\nabla^{2/3}\,C_T\,V^3}{\eta_D\,\eta_S} = \frac{\rho^{1/3}\,k_S\,C_T}{2\,\eta_D\,\eta_S}\;\Delta^{2/3}\,V^3$$

Substitute 2 into 1, with ∇ = Δ/ρ: brake power grows with Δ to the 2/3 and with V³

4

$$C = \frac{\Delta^{2/3}\,V^3}{P_B} = \frac{2\,\eta_D\,\eta_S}{\rho^{1/3}\,k_S\,C_T}$$

Hold CT, ηD, ηS, kS (same hull, Fn, Rn): C is constant (2.104), on PB

Symbols

R_T	total resistance [kN]
ρ	sea water density, 1.025 t/m³
S	wetted surface [m²]
V	speed [m/s]; [kn] in C
C_T	total resistance coefficient
P_E, P_B	effective, brake power [kW]
η_D, η_S	propulsive, shaft efficiency
∇	displaced volume [m³]
Δ	displacement [t]
C_S	Taylor coefficient, f(B/T, CM)
k_S	shape constant of the hull
C	Admiralty constant, typically 400–600 with Δ [t], V [kn], P [kW]

[1] Taylor’s formula estimates the wetted surface from ∇ and L alone, before any lines exist, for the frictional resistance; CS is read against B/T and CM (Lewis 1988). It suits conventional displacement hulls; for a bulb, transom or unusual sections use the lines, or the S of Holtrop–Mennen (Tool 2.4).

Power, Speed and Displacement: Three Relations

Power for a new speed or displacement

$$\frac{P_2}{P_1} = \left(\frac{\Delta_2}{\Delta_1}\right)^{2/3} \left(\frac{V_2}{V_1}\right)^{3}$$

Speed for a new power or displacement

$$\frac{V_2}{V_1} = \left(\frac{P_2}{P_1}\right)^{1/3} \left(\frac{\Delta_1}{\Delta_2}\right)^{2/9}$$

Displacement for a new power or speed

$$\frac{\Delta_2}{\Delta_1} = \left(\frac{P_2}{P_1}\right)^{3/2} \left(\frac{V_1}{V_2}\right)^{9/2}$$

Same ship, same C: write 2.104 for state 1 and state 2, and divide.

Change one by +10 %, hold another: the third follows

Hold	Change	Result
Δ	V +10 %	P +33.1 %
V	Δ +10 %	P +6.6 %
P	Δ +10 %	V −2.1 %
Δ	P +10 %	V +3.2 %
P	V +10 %	Δ −34.9 %

Speed is expensive, displacement is cheap: power goes with V³ but only with Δ to the 2/3.

The book's use (§1.3): C measured on sea trials at a reduced draught gives the speed at the design displacement with the installed power.

$$P_2 = P_1 \left(\frac{\nabla_2}{\nabla_1}\right)^{2/3}$$

The sea-trial standard uses the same law: ITTC 7.5-04-01-01.1, Speed/Power Trials (revision 08, 2024)

· §10.3.5 corrects the measured trial power for a different displacement, "based on the Admiralty formula", at the same speed

· §5.1 allows this only if the trial displacement is within 2 % of the required one

Order of Resistance, Order of Power, and the Humps

$$R \propto V^{\,n} \;\Rightarrow\; P = RV = aV^{\,m}, \qquad m = n + 1$$

$$C = \frac{\Delta^{2/3} V^3}{P} = \frac{\Delta^{2/3} V^3}{aV^{\,m}} = \frac{\Delta^{2/3}}{a} V^{\,3-m}$$

a: a constant of the ship. Δ fixed: C constant only if m = 3

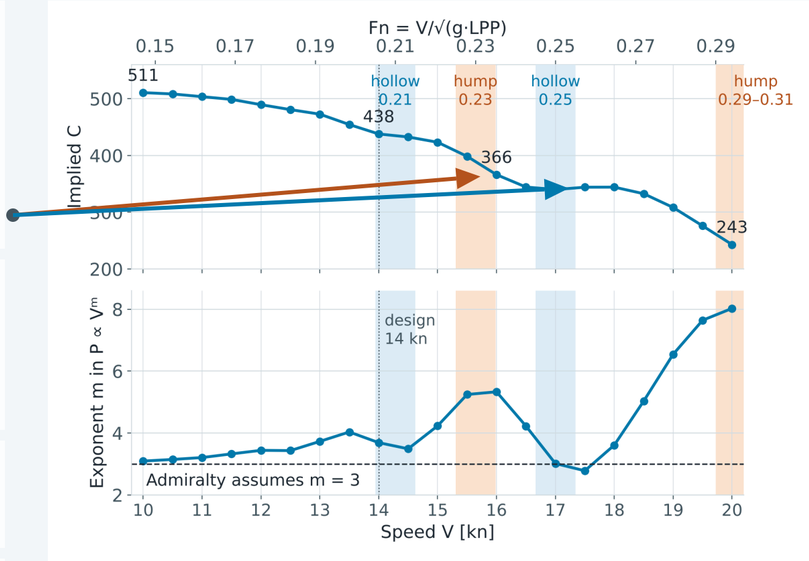
m > 3: C falls (hump) · **m < 3: C rises** (hollow)

$$C_F = \frac{0.072}{R_n^{0.2}} \;\Rightarrow\; R_F \propto V^{1.8}, \quad P_F \propto V^{2.8}$$

Friction alone (Prandtl, Eq. 2.9): m just below 3.

Waves: CW swings with Fn (Fig. 2.14). Towards a hump m rises above 3 and C falls; in a hollow m drops back to about 3.

Reference ship: m 3.1 at 10 kn; above 5 at 15.5–16 kn, on the 0.23 hump; 2.8 at 17.5 kn, in the 0.25 hollow. C falls from 511 to 243.



Computed, not measured: Holtrop–Mennen (Tool 2.4) on the reference ship, SMCR. A realistic estimate from a regression of model tests, not an idealized curve; above 16 kn this full hull is far from its design speed.

Range of validity, read from the chart: one C serves only where m ≈ 3

Same C across speeds? Within about ±1 kn

10–12 kn: 511 to 489 (−4 %) · 13.5–15 kn: 454 to 423 (±4 % of 438).

Not across a hump

15–16.5 kn, Fn 0.22–0.24: C drops 19 % (423 to 344); above Fn 0.27, again.

A hollow has its own C

16.5–18 kn, Fn 0.25: C flat at 340–344, but not the 438 of 14 kn.

A Sister Ship, Heavier and Faster

Question

The parent is the reference ship: Δ_0 = 13,200.7 t, SMCR P_0 = 3,500 kW at V_0 = 14 kn. A sister ship of the same hull form floats 5 % heavier, Δ = 13,860.7 t. **(a)** What SMCR does she need for 14.5 kn? **(b)** What speed does she make on the parent's 3,500 kW? **(c)** The owner then asks for 16 kn: can the same C be used?

C $$C = \frac{\Delta_0^{2/3} V_0^3}{P_0} = \frac{13\,200.7^{2/3} \cdot 14^3}{3\,500} = 438$$ The parent's constant, on the SMCR basis. Same hull form, so it carries over.

(a) $$P = P_0 \left(\frac{\Delta}{\Delta_0}\right)^{2/3} \left(\frac{V}{V_0}\right)^3 = 3\,500 \cdot 1.05^{2/3} \cdot \left(\frac{14.5}{14}\right)^3 = 4\,017\ \text{kW}$$ **Check:** Holtrop–Mennen 4,065 kW, 1 % apart. Valid: Fn 0.213, 0.5 kn from the parent.

(b) $$V = V_0 \left(\frac{\Delta_0}{\Delta}\right)^{2/9} = 14 \cdot 1.05^{-2/9} = 13.85\ \text{kn}$$ **Check:** Holtrop–Mennen 13.88 kn. Speed is a weak function of displacement.

(c) $$P = 3\,500 \cdot 1.05^{2/3} \cdot \left(\frac{16}{14}\right)^3 = 5\,397\ \text{kW}$$ **Check:** Holtrop–Mennen 6,476 kW: 17 % short. Fn 0.235 is on the 0.23 hump (slide 4): **not valid**, recompute.

Answer: (a) 4,017 kW · (b) 13.85 kn · (c) no: across a hump the answer is 17 % short. Use C within about a knot.